Applications of Principal Component Analysis (PCA)

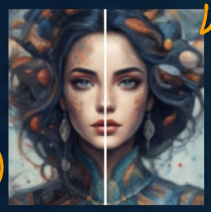
Image Compression

Original High-Res Image

Original High-Res Image

PCA reduces file sizes while keeping essential features

Compressed (PCA)



Compressed (PCA)

Data Visualization

High-Dimensional Data
(e.g., 50 features)

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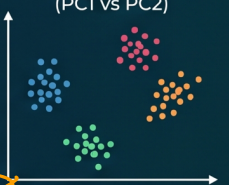
(e.g., 50 features)

Complex datase with dozens of variables to 2D easy viewing.

2D Visualization
(PC1 vs PC2)



2D Visualization
(PC1 vs PC2)

Facial Recognition (Eigenfaces)

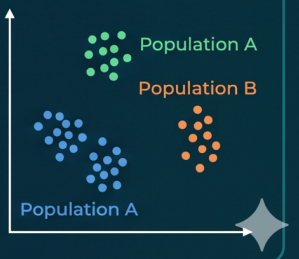
Based on facial datasets

PCA **isolates** the **most prominent** facial features (like eyes and nose shape) from orlesrties, etc.)

PCA isolats the most **prominent** facial features algorithme for security or identification algorithms

Genetics & Biological Data

PCA uncovers **patterns**, **ancestry**, genetic diversity grouping thousands of genes into core factors



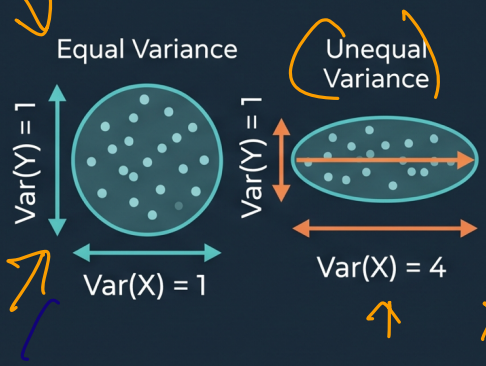
1. Covariance Matrix
2. Single Value Decomposition (SVD)
3. Random Projection

Relative variance of features

What Does a Covariance Matrix Mean Visually?

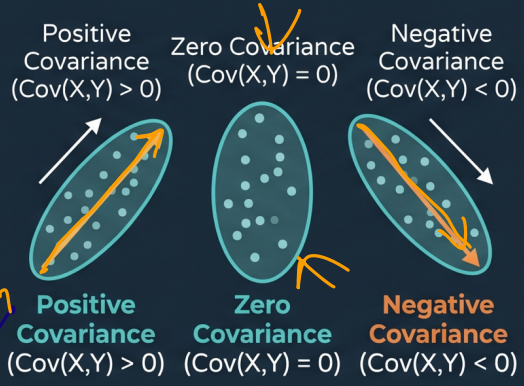
The covariance matrix acts as a geometric blueprint for a dataset's shape, spread, and tilt.

1. Diagonal Entries: Variance (Spread)



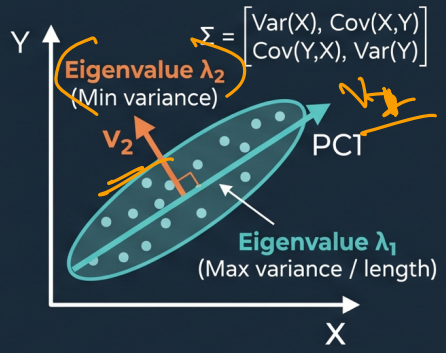
Determine the width and height of the data cloud along the original X and Y axes.

2. Off-Diagonal Entries: Covariance (Tilt)



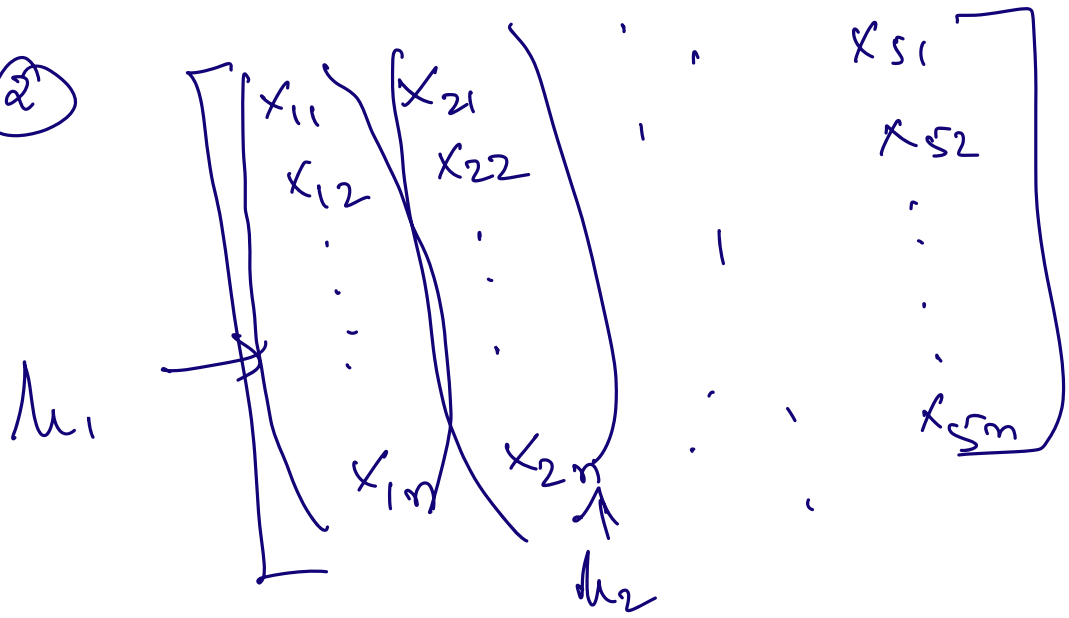
Determine the direction and strength of the tilt (correlation) between the variables.

3. Geometric Interpretation



The eigenvectors (v) point along the axes of the ellipse, while eigenvalues (λ) show how far the stretches along those directions.

① We have a dataset which has 5 features n samples

② 

③ Compute means $\mu_1, \mu_2, \cdots, \mu_5$

④
$$\begin{bmatrix} x_{11}-\mu_1 & x_{21}-\mu_2 & \cdots & x_{51}-\mu_5 \\ x_{12}-\mu_1 & x_{22}-\mu_2 & & x_{52}-\mu_5 \\ \vdots & \vdots & & \vdots \\ x_{1n}-\mu_1 & x_{2n}-\mu_2 & \cdots & x_{5n}-\mu_5 \end{bmatrix}$$

5) $Cov = \frac{1}{n-1}(X-\bar{x})^T(X-\bar{x})$

↑ Square matrix → 5×5 → Symmetric

$$\begin{bmatrix} \sigma_1 & Cov_{21} & \cdots & \cdots \\ Cov_{12} & \sigma_2 & \ddots & \vdots \\ Cov_{13} & \ddots & \ddots & \vdots \\ Cov_{14} & \cdots & \cdots & \ddots \\ Cov_{15} & \cdots & \cdots & \cdots \end{bmatrix}$$

6) Eigen values, Eigen vectors of the covariance matrix

7)

F.Val → λ_1 E.Vect → v_1

$\lambda_2 v_2$

$\lambda_3 v_3$

$\lambda_4 v_4$

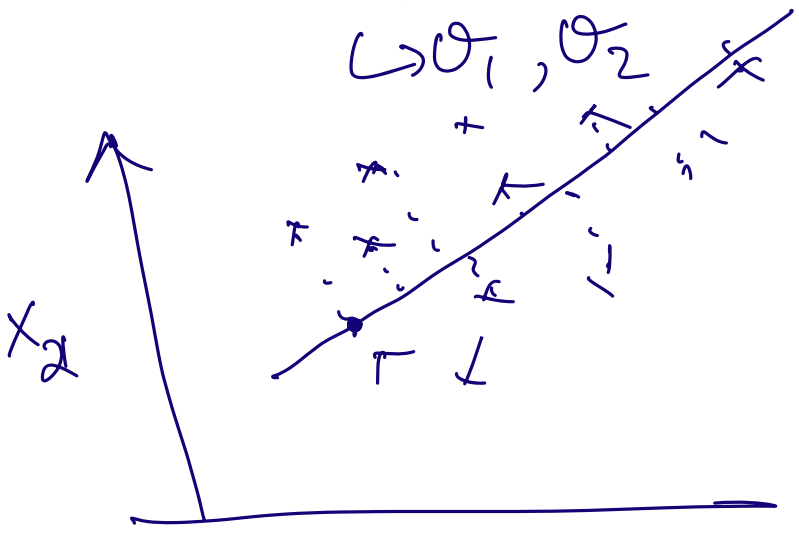
$\lambda_5 v_5$

High e.val

Ascending order

small E.val

(7) Projection Matrix using the chosen Eigen Vectors



$$P = \begin{bmatrix} \frac{\theta_1}{\|\theta_1\|_2} & \frac{\theta_2}{\|\theta_2\|} \end{bmatrix}^T$$

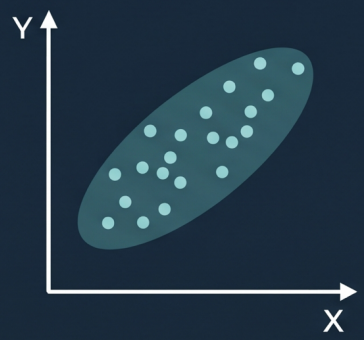
(7) $X_{PCA} = \underline{(X - \mu) \cdot V}$

$\uparrow$
reduced

The Covariance-Based Approach to PCA

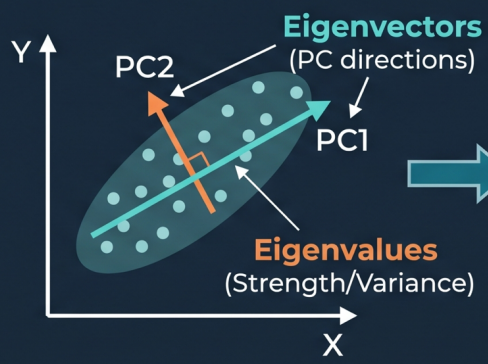
How we find the directions of maximum variance using data relationships.

1. Calculate Covariance Matrix



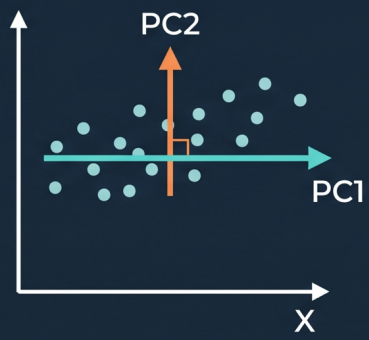
$$\Sigma = \begin{pmatrix} \text{Cov}(X,X) & \text{Cov}(X,Y) \\ \text{Cov}(Y,X) & \text{Cov}(Y,Y) \end{pmatrix}$$

2. Eigenvalue Decomposition



$$\Sigma \mathbf{v} = \lambda \mathbf{v}$$

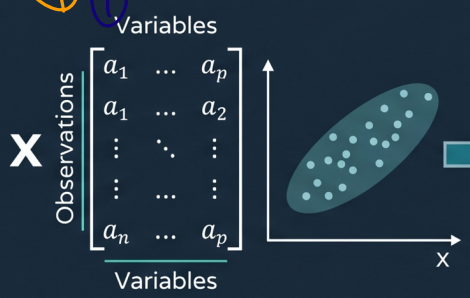
3. Project Onto Principal Components



The SVD Approach to Principal Components

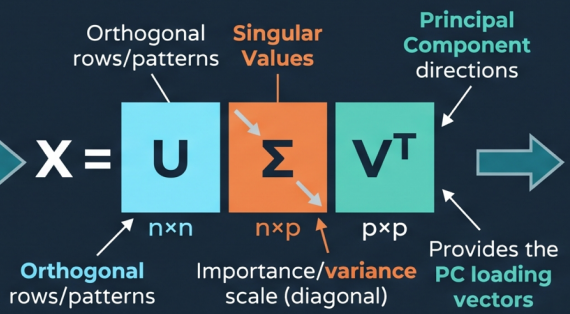
How Singular Value Decomposition decomposes data to find PCs without calculating a covariance matrix.

1. The Data Matrix (X)



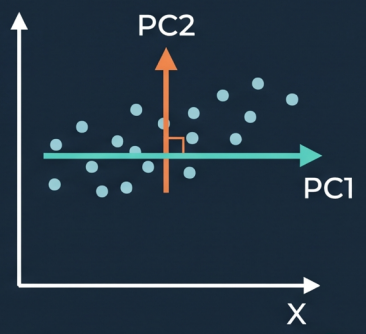
Start with the original centered data matrix **X** (n × p). SVD bypasses the need for the n × n covariance matrix, making it computationally efficient for large datasets.

2. Singular Value Decomposition



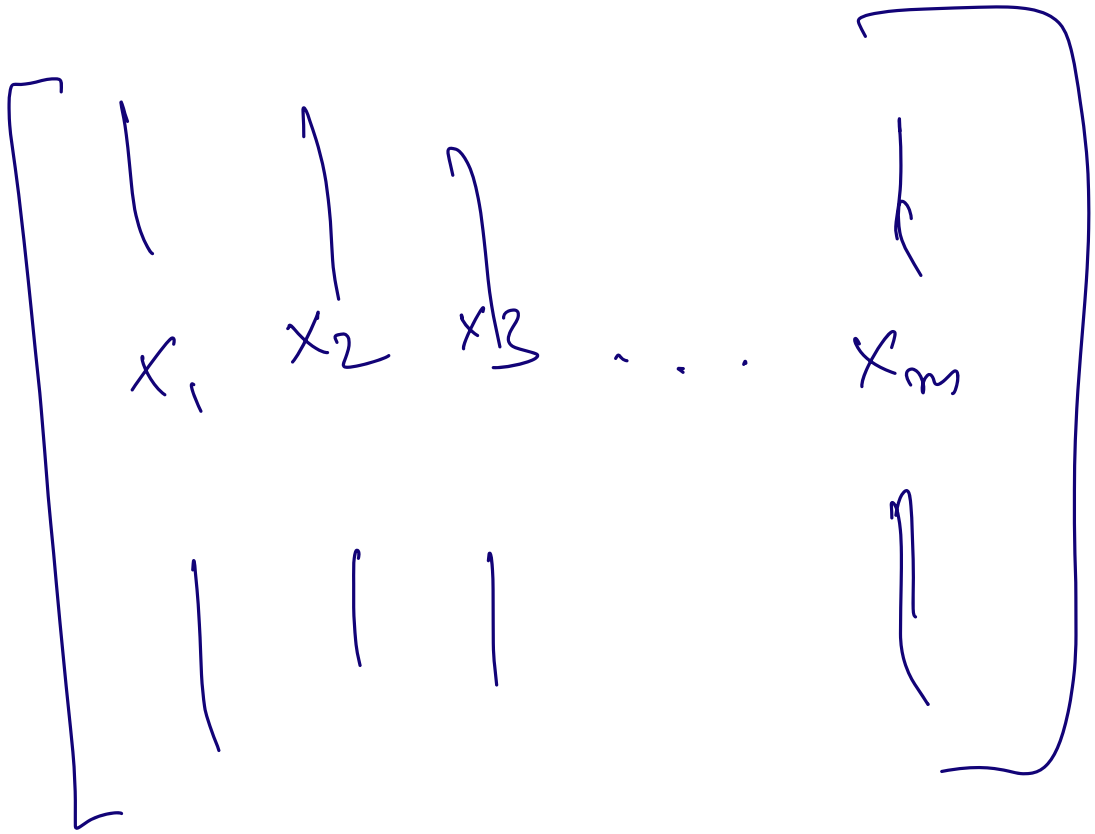
Decomposing **X** directly into patterns (**U**), scaling factors (**Σ**), and direction vectors (V^T).

3. Projecting onto Principal Components

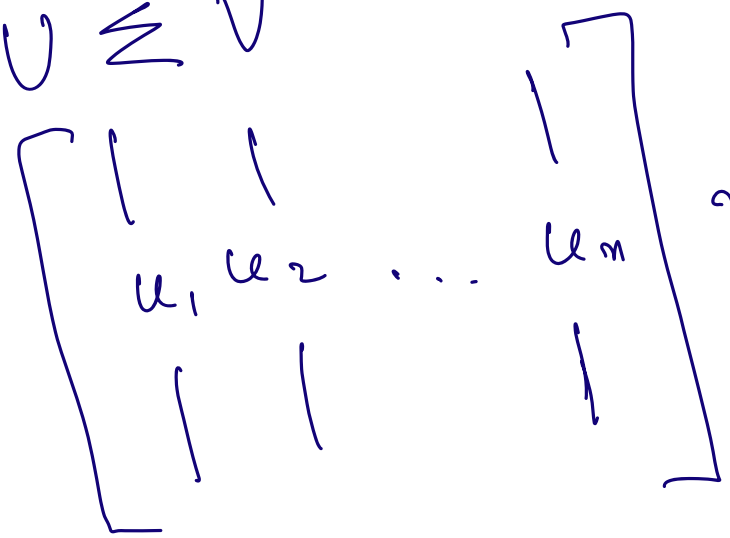


The right singular vectors (V^T) define the new axes. The product **T = UΣ** (or **XV**) projects the original data into the principal component space.

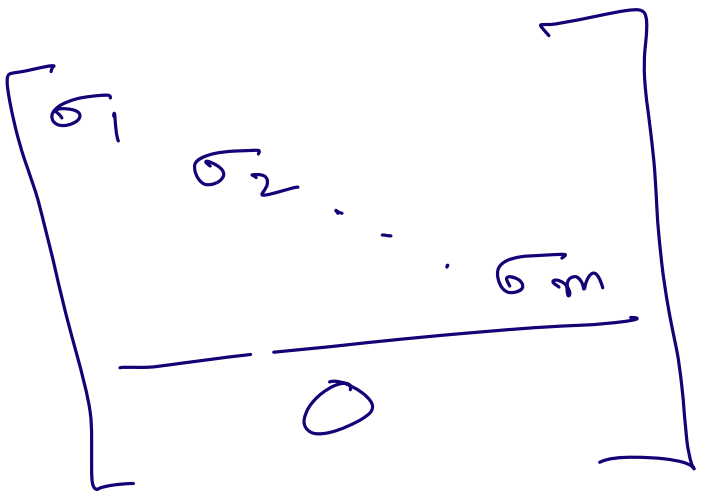
$$X = U \Sigma V^T$$

X_2 

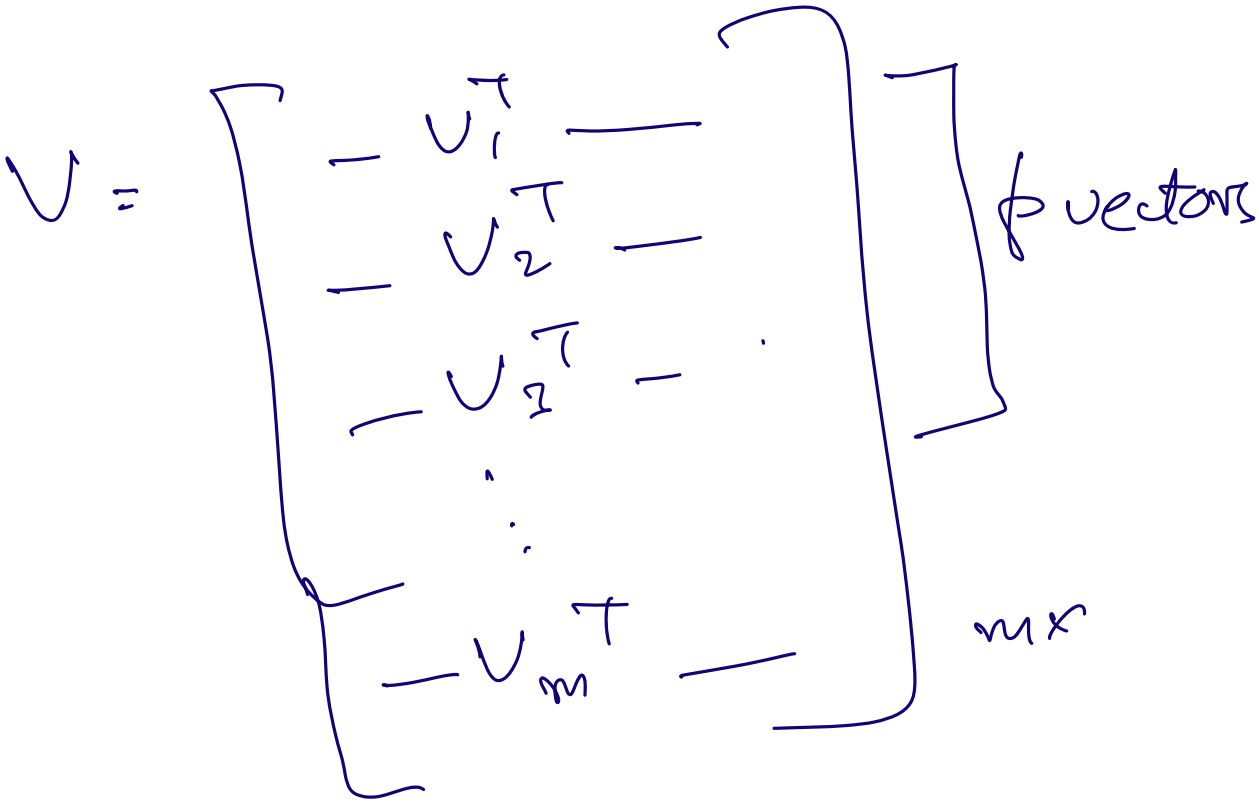
$\rightarrow U \Sigma V^T$

$U =$ 

$n \times n$

Σ_2 

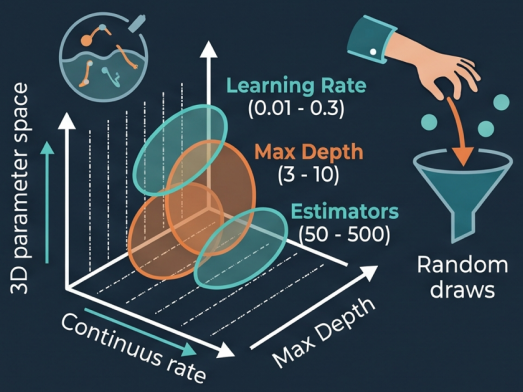
$m \times m$ diagonal matrix variance along the PC.


V =
$-v_1^T-$
$-v_2^T-$
$-v_3^T-$
$-v_m^T-$
vectors
m x

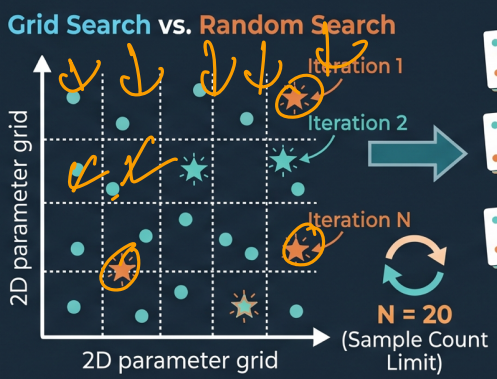
How RandomizedSearchCV Works

Efficient hyperparameter tuning by randomly sampling from a defined search space instead of testing every combination.

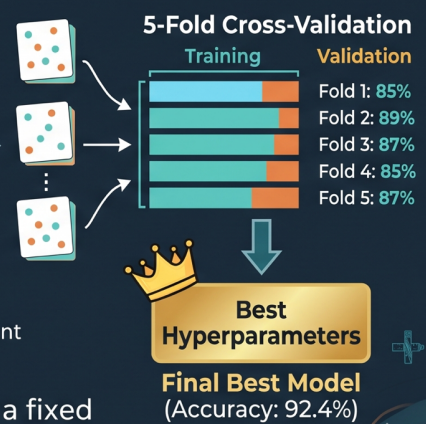
1. Define Search Space



2. Random Sampling (N Iterations)



3. Evaluate & Pick Best



Unlike Exhaustive Grid Search, RandomizedSearchCV evaluates a fixed number of random configurations (N iterations), saving vast amounts of computation time while finding near-optimal hyperparameters.

Based on Johnson & Lindenstrauss lemma

say samples $m = 5000$

$n = 20000$

If you donot want squared distance b/w any (two points to change more) than $\epsilon = 10\%$, then

$$d \geq 4\log(m) / \left(\frac{1}{2}\epsilon^2 - \frac{1}{3}\epsilon^3\right)$$

Random matrix $P[d, n]$ where each item in the matrix is sampled randomly from a Gaussian distribution

$$X_reduced = X \underline{@} P.T$$

Choose PCA

1. moderate no. of feature: few hundred few thousand
2. Max info retention
3. PCA filters random noise indata

Choose Random Projection

① very large no:of features
→ NLP with large vocab
→ genome sequencing

② PCA very slow → As no:of features goes
compute complexity
is increasing cubic
time

③ KNN, clustering